

Alejandro Vaz

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Experience

SmallVec (Rust)

03/26 – Present

Open Source Contributor

Identified and fixed the causes of the performance regressions introduced in the upstream Rust crate SmallVec (~1B downloads) v2 update and microbenchmarked the improvement.

- Improved extend performance by +140%.
- Improved push performance by +30%.
- Improved pushpop performance by +40%.
- Submitted upstream [Pull Request](#).

Independent Research

02/26 – 07/26

Independent Researcher

Developed a novel contextual disambiguation algorithm for generalized parsers and formalized it in a paper.

- Pre-print available on [arXiv](#) and [Zenodo](#).
- Submitted to the Software Language Engineering 2026 conference.
- Implemented along the Mathsys Earley parser.
- Name: "Environment-Sensitive Lexicographic Disambiguation for Contextual Parsing".

Rust Stack-Array

06/26 – Present

Crate maintainer

Owner of the Rust crate [Stack-Array](#) with ~20k downloads.

- Assumed ownership of the crate after developing a new implementation with compile-time evaluation support replacing the original codebase.
- Expanded the public API with 15+ operations, iterators and traits.

Projects

Mathsys

07/25 – Present

[Mathsys](#) is Domain-Specific Language (DSL) for simplified mathematical typesetting.

- Hand-rolled lexer/tokenizer (zero-copy byte-scanner).
- Hand-rolled recursive descent parser (backtracking).

Skills

Languages: Rust, Python, TypeScript, Shell.

Tools: Git, Linux, Cargo.

Areas: lexing, parsing, performance, algorithms, unsafe Rust.

Education

Colegio Mater Immaculata / IES Lope de Vega

09/24 – 05/26

Bachillerato Tecnológico - 8.89/10 GPA

Founded the school's programming club which reached 34 members across five grade levels.

Universidad Complutense de Madrid

Starting September 2026

Double Degree on Business and Computer Science